

SALES CASE STUDY

SALES PERFORMANCE, PRICING,
PROFITABILITY & PRICE ELASTICITY
ANALYSIS

Tools: Databricks | SQL | Power BI | Lovable
| GitHub

NTSHOVELO
HLONGWANE
BRIGHT LEARN

Year: 2026



Business Problem

This analysis uses daily sales data to uncover the relationship between pricing, volume, and profitability — with a focus on promotional impact.

Sales Case Study

BUSINESS OBJECTIVE

Understand sales performance, pricing, and profitability over time
Analyse the impact of promotional pricing on volume and margin
Identify pricing inefficiencies and opportunities for improvement
Support data-driven pricing and promotional strategy decisions

KEY QUESTIONS

What is the daily sales price per unit over time?
How profitable are sales — and on how many days was gross profit negative?
How does quantity sold respond to changes in price?
What impact do promotions have on volume and gross profit?
Do lower prices generate enough volume to justify price cuts?
What pricing and promotional strategies could improve profitability?

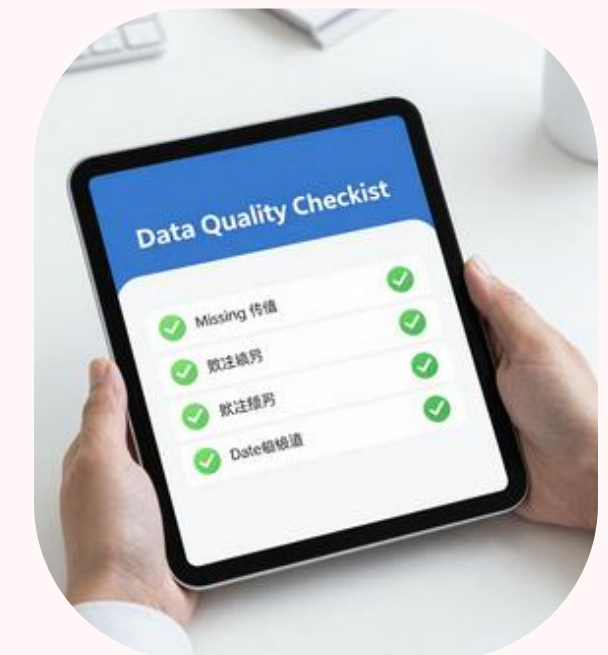
Dataset & Data Preparation

The dataset contains 1,053 records spanning 30 Dec 2013 to 16 Nov 2016. Raw variables include Date, Sales, Cost of Sales, and Quantity Sold. Data preparation involved checking for missing values, duplicates, and invalid entries, converting date formats, and creating calculated sales and profitability metrics.

Data Records

1,053

Records retained with no missing or duplicate dates ✓



Calculated Metrics

To enable a comprehensive analysis of sales performance and profitability, several key metrics were derived from the raw dataset. These calculated fields allow assessment across four dimensions: revenue, volume, profitability, and pricing efficiency — forming the analytical foundation of this case study.



Sales Price per Unit = Sales
÷ Qty Sold
Cost per Unit = Cost of
Sales ÷ Qty Sold



Gross Profit = Sales - Cost
of Sales
Gross Profit % = (GP ÷
Sales) × 100



GP per Unit = Gross Profit ÷
Qty Sold
GP % per Unit = (GP per Unit
÷ Price) × 100

Overall Performance

Total Sales

R186.91M

Total Quantity Sold: 5.28M units |
Cost of Sales: R194.03M | Avg Daily
Sales: R177.50K

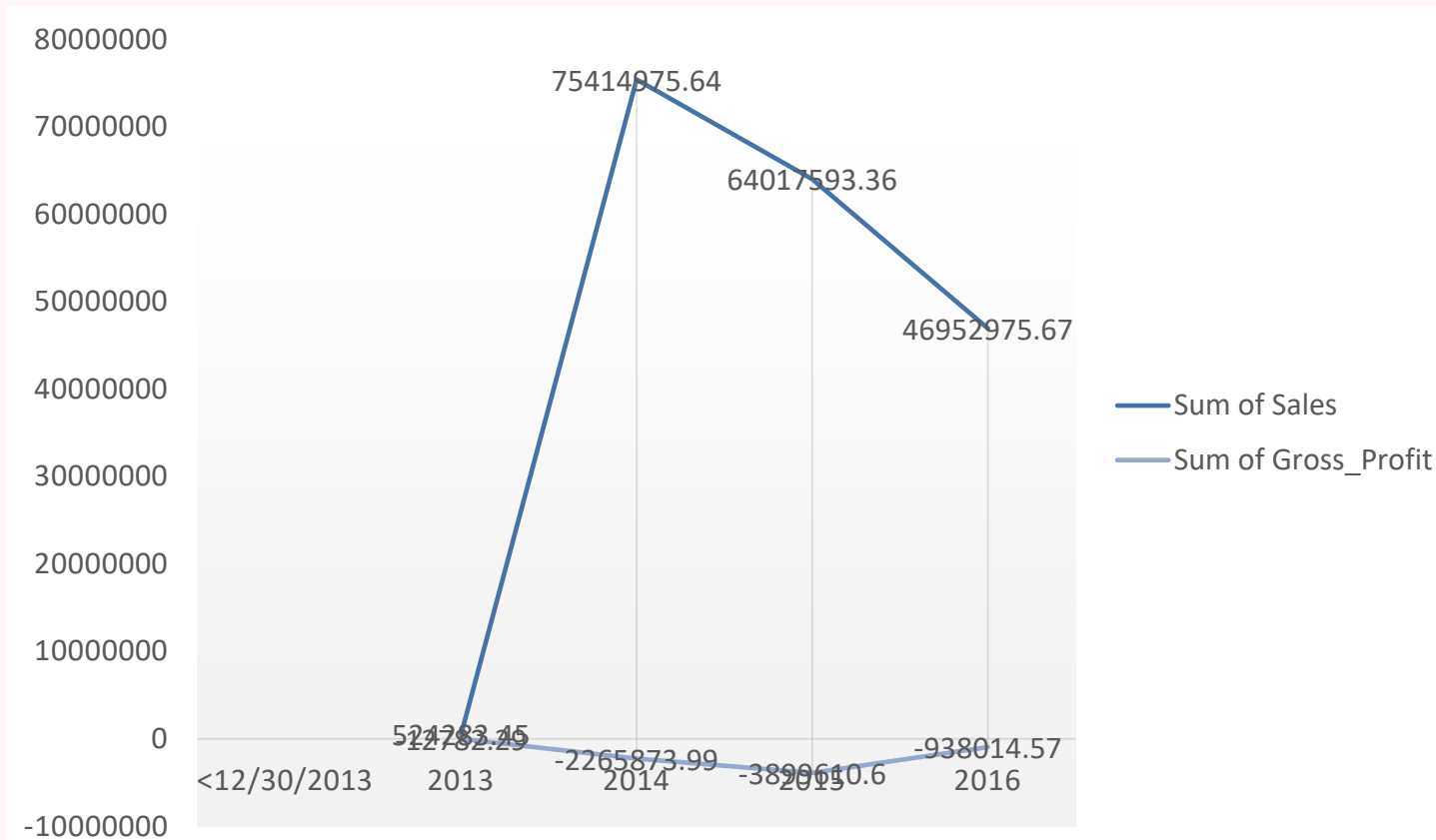
-R7.12M *Gross Profit*

Gross Profit Margin: -3.81% | Despite strong volume, cost exceeded revenue causing a gross loss

R35.40 *Avg Price/Unit*

607 of 1,053 days (57.6%) recorded negative gross profit — profitability was not the norm

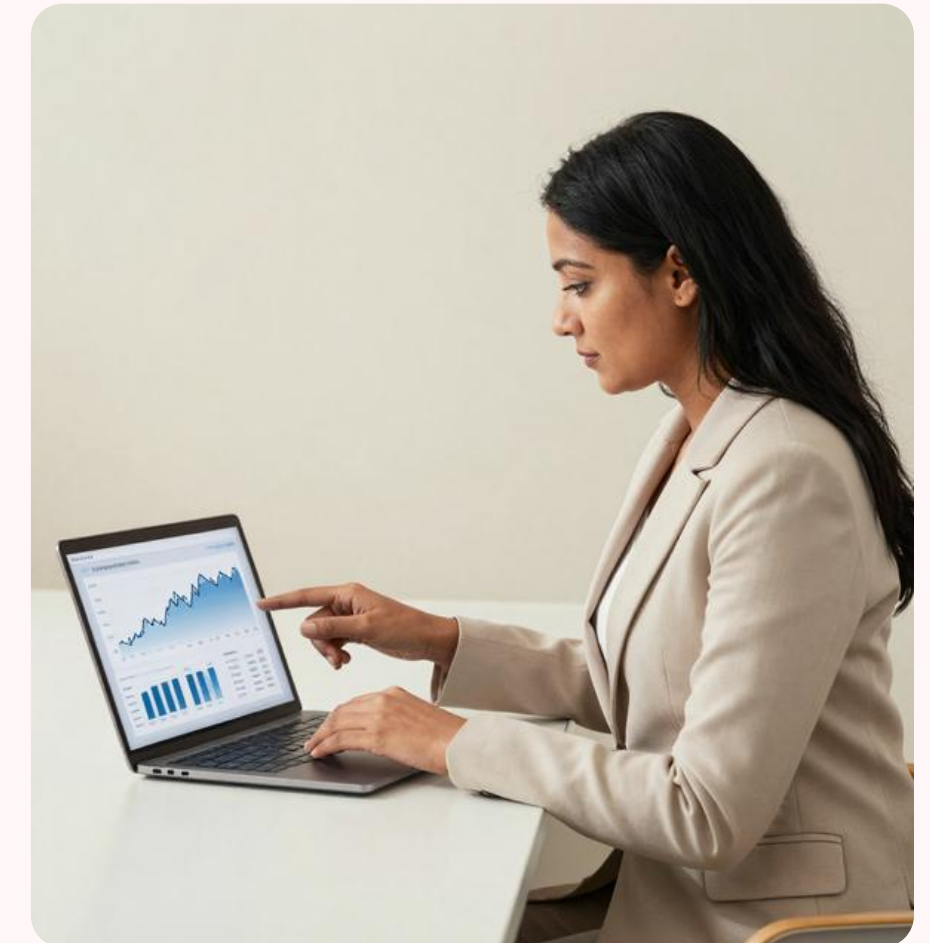




Sales varied considerably over time. Some high-volume days generated strong revenue but still recorded negative gross profit, revealing a disconnect between sales volume and profitability.

PEAK SALES DAY

1 March 2014: R846.68K in sales, 26,410 units sold — yet gross profit was -R25.4K



KEY INSIGHT

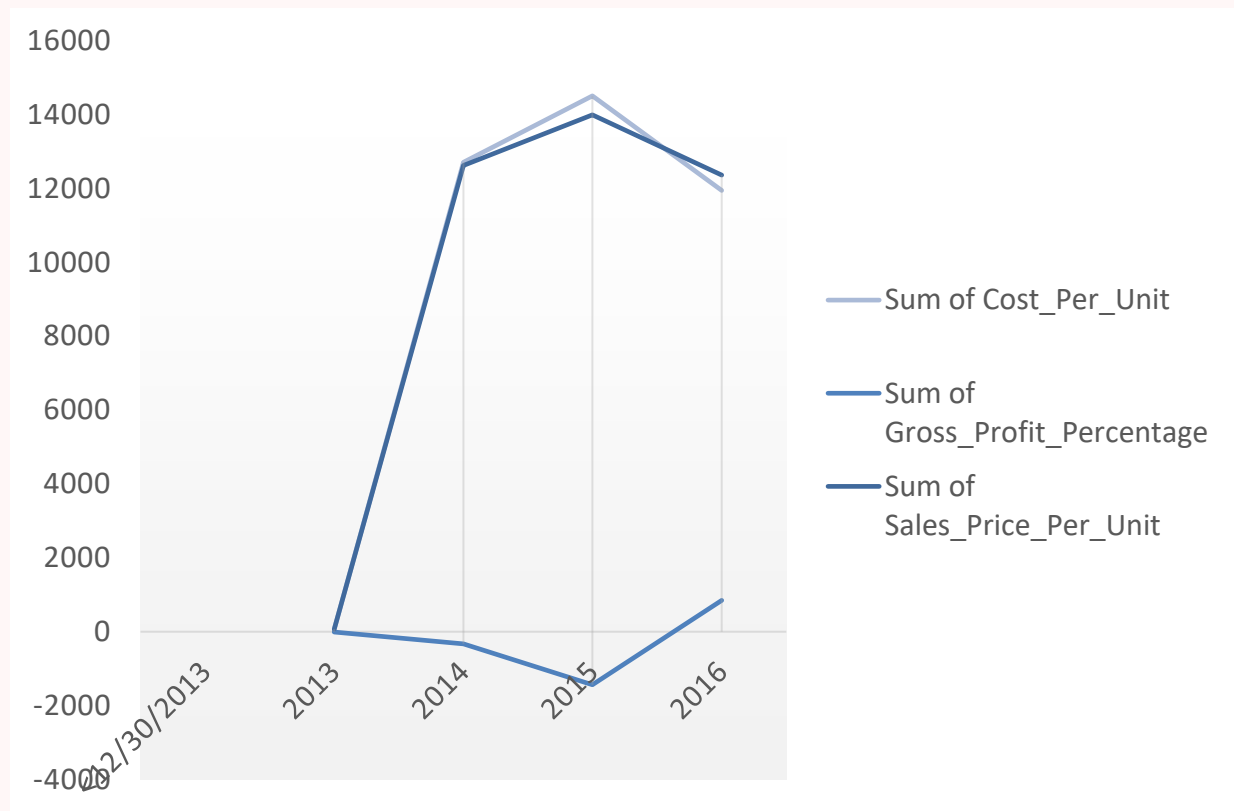
"The highest sales day was not necessarily the most profitable day."

GROSS PROFIT TREND

Gross profit fluctuated significantly, with 57.6% of days (607 of 1,053) recording negative gross profit margins

Sales & Profitability Trends

Pricing & Profitability Trends



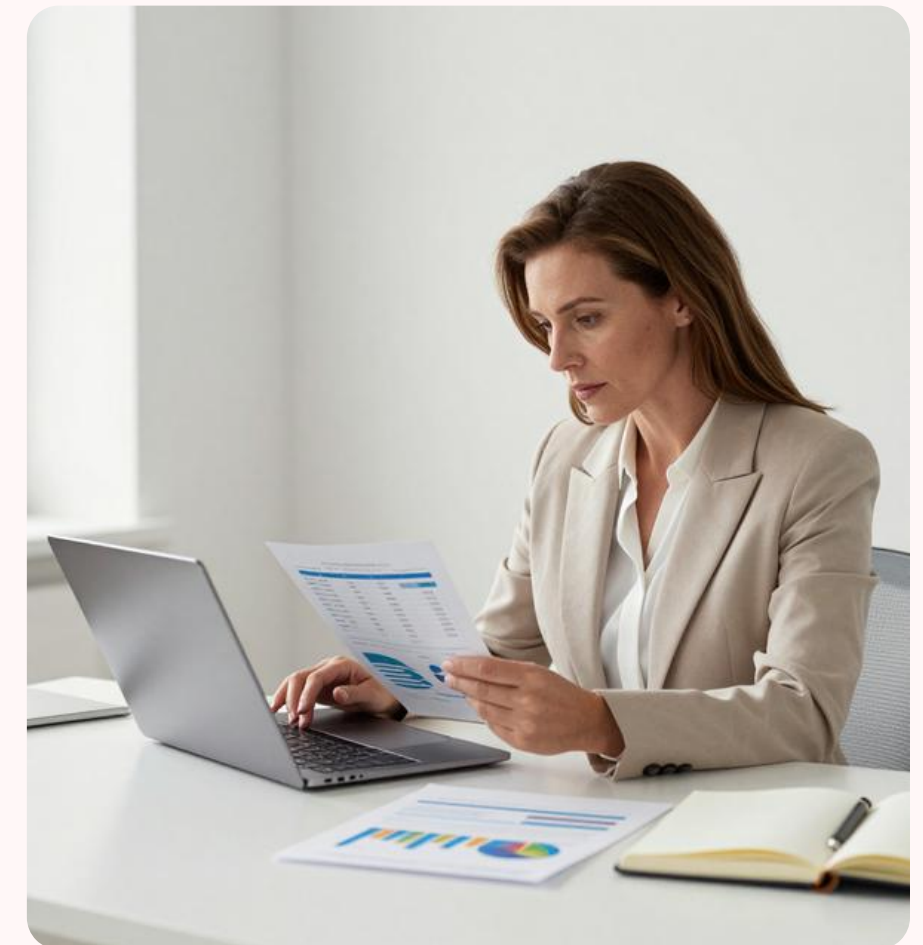
Price per unit increased over the period while cost per unit remained elevated. Higher prices in 2016 finally aligned with improved gross profit margins, reversing prior years of negative profitability.

2014

Avg. Price: R34.57/unit
Gross Profit Margin: -0.90%
Price below cost threshold

2015

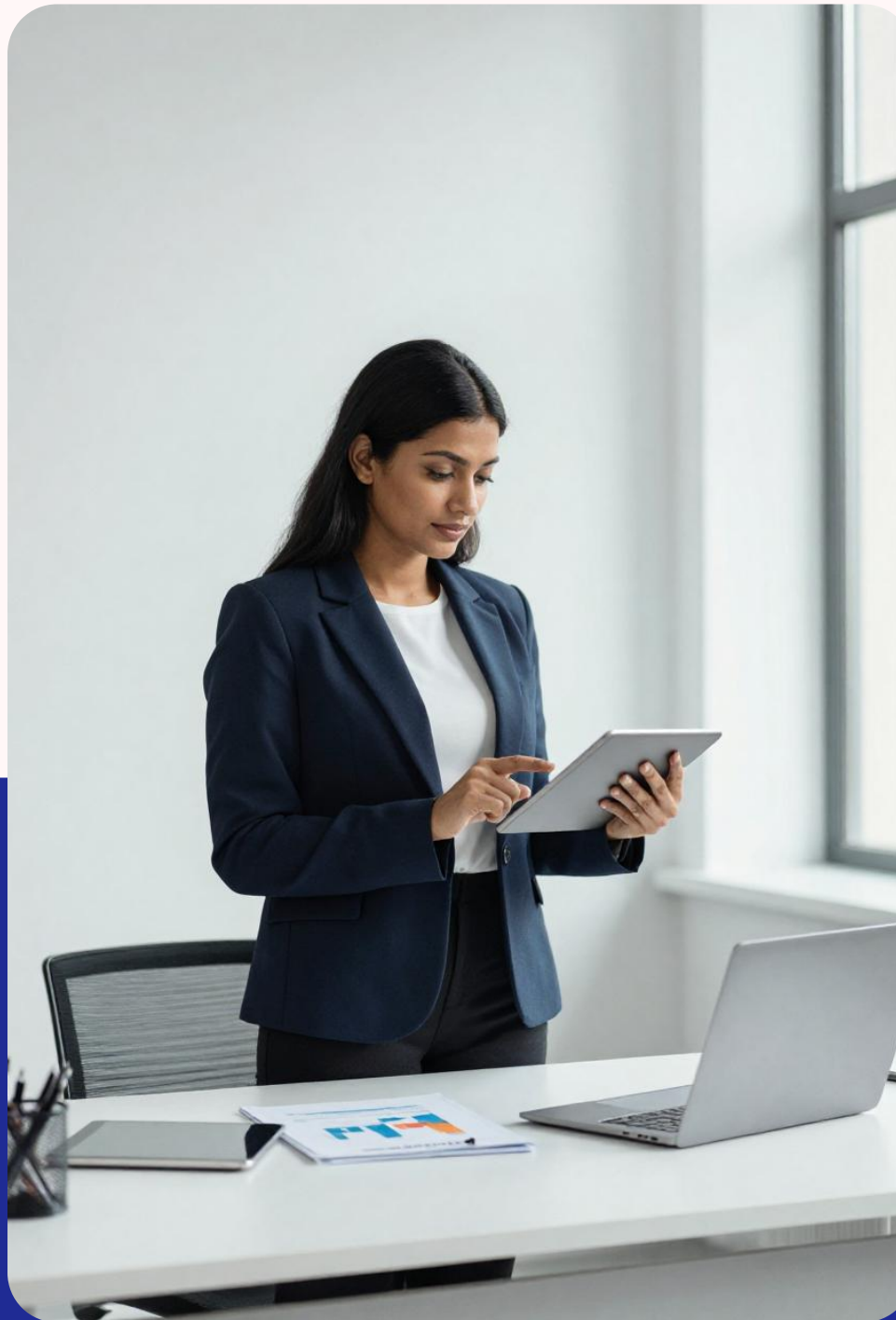
Avg. Price: R38.33/unit
Gross Profit Margin: -3.93%
Volume-driven losses persisted



2016

Avg. Price: R38.52/unit
Gross Profit Margin: +2.64%
First year of positive margin

Impact of Promotional Pricing



Sales Performance

2029

-14.65%

Largest price cut (Promo 2) drove +503.9% quantity increase — yet still negative gross margin

Promo 1 (Aug–Sep 2014): -4.35% price, +57.74% qty, elasticity -9.29, negative margin.

Promo 2 (Sep–Oct 2015): -14.65% price, +503.94% qty, elasticity -34.41, negative margin.

Promo 3 (Jun–Jul 2016): -6.30% price, +64.8% qty, elasticity -14.90, negative margin.

All promotional periods resulted in negative gross profit.

Promotion Insights

Promotions successfully stimulated demand across all three periods. While each promotion drove significant volume increases, the additional units sold did not compensate for the reduction in unit economics. All promotional periods resulted in negative gross margins, raising questions about the true value of discount-driven strategies.

+503.9%

Quantity surge in Promo 2 after a 14.65% price cut — the largest volume response observed

Promo 1: +57.7% | Promo 2: +503.9%
| Promo 3: +64.8% quantity increase

Key Insight

Higher volume did not improve profitability — all promos recorded negative gross profit and negative margins

Promotion Insights

2029



Price Elasticity

Price Elasticity = % Change in Quantity Sold ÷ % Change in Price. All three promotional periods produced elasticity values negative and greater than 1 in absolute value, indicating strong consumer responsiveness to price reductions. Caveat: these are observational estimates and do not prove that price alone caused demand changes — other factors may have contributed.



Promo 1 Elasticity: -9.29
Price cut -4.35%,
volume +57.74%



Promo 2 Elasticity: -34.41
Price cut -14.65%,
volume +503.94%



Promo 3 Elasticity: -14.90
Price cut -4.50%,
volume +64.8%

Key Insights

Insight 1

Revenue $\neq$ Profit

R186.91M in sales yet -R7.12M gross profit.
Strong volume alone does not guarantee profitability.

Price & Volume

Insight 2

Price has a strong relationship with volume. Promotions led to substantial quantity increases of up to +503.9%.

Margin Impact

Insights 3 & 4

All promos had negative margins despite volume gains.
Higher pricing in 2016 improved margin to +2.64% vs. -3.93% in 2015.



Recommendations

Translating key findings into actionable pricing and promotional decisions to improve profitability and long-term sales strategy.

CHALLENGES

All three promotional periods resulted in negative gross profit margins despite significant volume increases
Decision-making has been driven by volume and revenue rather than incremental gross profit impact
Price elasticity estimates are observational and cannot be treated as definitive proof that price alone drives demand changes
Current pricing approach lacks a defined minimum profitable selling price to protect unit economics

OPPORTUNITIES

Review promotional pricing by evaluating incremental gross profit, not just volume or revenue generated
Establish a minimum profitable selling price (pricing floor) to safeguard margins on every transaction
Use targeted promotions with smaller discounts and measure responses across quantity, revenue, profit, and customer metrics
Monitor price elasticity in future pricing decisions rather than relying solely on historical volume trends



Conclusion

Sales volume alone does not guarantee profitability. Promotional pricing strongly increases quantity sold but consistently caused negative gross margins across all three promotional periods analyzed. Balancing price, volume, and margin is critical for sustainable sales and promotional strategy success.

-3.81%

Overall gross profit margin across the full analysis period

Key Opportunity:
Identify pricing and promotional levels that increase demand while maintaining positive unit economics.

THANK YOU!

Sales Performance, Pricing, Profitability & Price Elasticity Analysis

